RIZVI COLLEGE OF ENGINEERING

Department of Artificial Intelligence & Data Science

NLP ASSIGNMENT – 1

E-Governance Feedback Analysis

Sentiment Analysis and Marathi-English Translation

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Abstract

E-governance platforms receive citizen feedback about public services, administrative delays, and unresolved applications. Natural Language Processing (NLP) can help organize such feedback by identifying sentiment, extracting key information, and grouping similar complaints. This assignment studies E-Governance Feedback Analysis using CPGRAMS as a real-world application context. A publicly accessible sentiment-analysis demonstration and Google Translate were tested using fictional English and Marathi feedback. The six sentiment examples covered positive, negative, and mixed feedback; translation tests examined whether the main meaning was preserved. The sentiment model identified clear positive and negative examples but assigned a single dominant label to mixed feedback. The tested translations retained the main meaning in these examples. The report proposes Marathi adaptation steps involving domain-specific data, Devanagari processing, dialect coverage, code-switching, and evaluation. The experiment is small and does not evaluate CPGRAMS's internal analytical system.

Keywords: NLP, E-Governance, Grievance Analysis, Sentiment Analysis, Marathi, Machine Translation.

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1. Introduction

1.1 Background

E-governance uses information and communication technologies to deliver public services and receive citizen feedback. Online grievance portals allow citizens to report problems involving certificates, ration cards, utilities, and other government services. Reviewing large volumes of natural-language complaints manually can be time-consuming.

NLP enables computers to process human language. In grievance management, it can support sentiment classification, complaint categorization, keyword extraction, duplicate detection, and topic identification.

1.2 Problem Statement

Feedback may be written in English, Marathi, or a mixture of languages. A single sentence can praise one aspect of a service while criticizing another. A system that assigns only one sentiment label may fail to represent this contrast. Translation can also be affected by regional expressions, informal spelling, and administrative terminology.

1.3 Objectives

1. Study E-Governance Feedback Analysis as an NLP application.
2. Test sentiment analysis on English and Marathi examples.
3. Examine Google Translate outputs for meaning preservation.
4. Identify observed limitations and possible error types.
5. Recommend practical steps for Marathi adaptation.

1.4 Scope

Six fictional sentences were tested: three English and three Marathi. CPGRAMS is the real-world application context; its restricted internal analytical system was not directly tested. Findings therefore apply only to the accessible tools and the small sample set.

2. Tools and Methodology

2.1 Real-World Application: CPGRAMS

The Centralised Public Grievance Redress and Monitoring System (CPGRAMS) is a Government of India platform for lodging and tracking public grievances. Official website: https://pgportal.gov.in/. It provides the application context because citizen complaints contain natural-language descriptions of service-related problems. The internal analytical dashboard was not accessed.

2.2 Tool A: Sentiment Analysis

A public sentiment-analysis demo classified text as positive, negative, or neutral and displayed scores. **Model page:** https://huggingface.co/Horizon-Labs/multilingual-sentiment-small Scores were transcribed from the experiment screenshots. They are not independently calibrated probabilities.

2.3 Tool B: Google Translate

Google Translate was used for Marathi-English and English-Marathi translation. Website: https://translate.google.com/. Translation was evaluated by comparing the output with the original sentence's main meaning. Since the tools perform different tasks, their outputs are not compared using one common accuracy score.

2.4 Dataset and Procedure

The six fictional samples represented negative, positive, and mixed feedback. Each was entered into the sentiment demo and the displayed label and scores were recorded. Marathi samples were translated into English. One additional mixed-sentiment English sentence was translated into Marathi. No real citizen data was used.

Table 2.1: Test sample overview

ID	Language	Intended content
E1	English	Negative: income certificate delayed
E2	English	Positive: ration card issue resolved quickly
E3	English	Mixed: usable portal, delayed status update
M1	Marathi	Negative: income certificate delayed
M2	Marathi	Positive: ration card issue resolved quickly
M3	Marathi	Mixed: usable portal, delayed status update

3. Part I: Real-World Tool Testing

3.1 Sentiment Analysis Results

Table 3.1 records the observed outputs from the supplied screenshots.

Table 3.1: Observed sentiment-analysis results

ID	Language	Label	Neg.	Neu.	Pos.
E1	English	Negative	100%	0%	0%
E2	English	Positive	0%	0%	100%
E3	English	Neutral	18%	80%	1%
M1	Marathi	Negative	96%	4%	0%
M2	Marathi	Positive	0%	0%	100%
M3	Marathi	Negative	70%	28%	2%

The E3 displayed scores total 99%, likely due to display rounding or precision.

3.2 English Examples

E1 input: "I submitted my income certificate application two months ago, but it is still pending. The delay is unacceptable." The model returned negative (100%), agreeing with the intended sentiment.

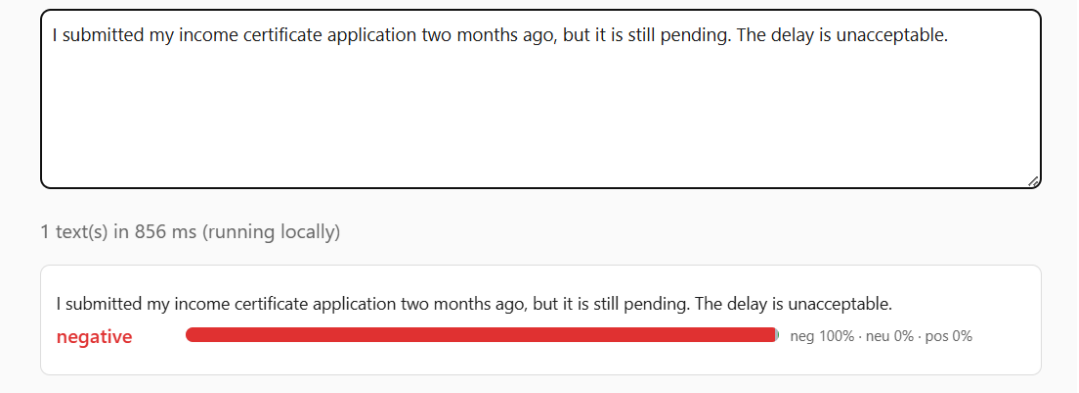


Figure 3.1: Sentiment-analysis result for English sample E1.

E2 input: "The officer resolved my ration card issue quickly. Thank you for the helpful service." The model returned positive (100%), agreeing with the expression of appreciation.

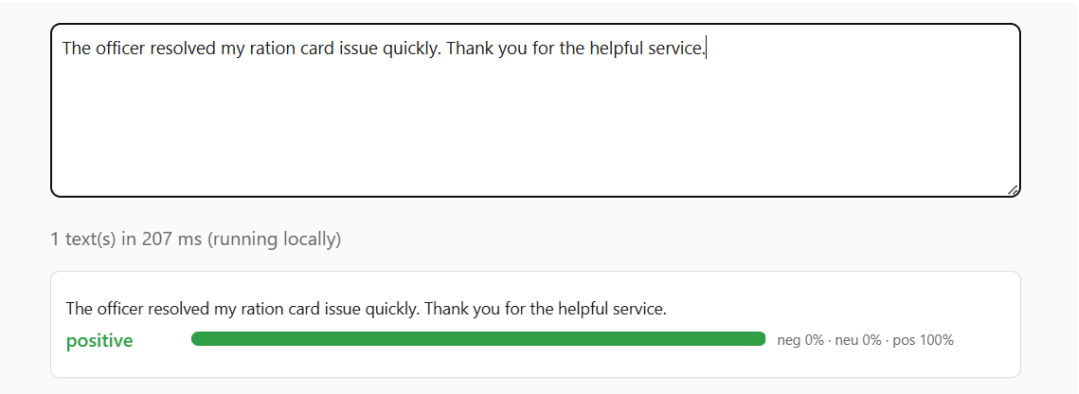


Figure 3.2: Sentiment-analysis result for English sample E2.

E3 input: "The online portal is easy to use, but the application status has not been updated for three weeks." The model returned neutral (80%). The sentence combines praise and dissatisfaction; one overall label does not explicitly represent both aspects.

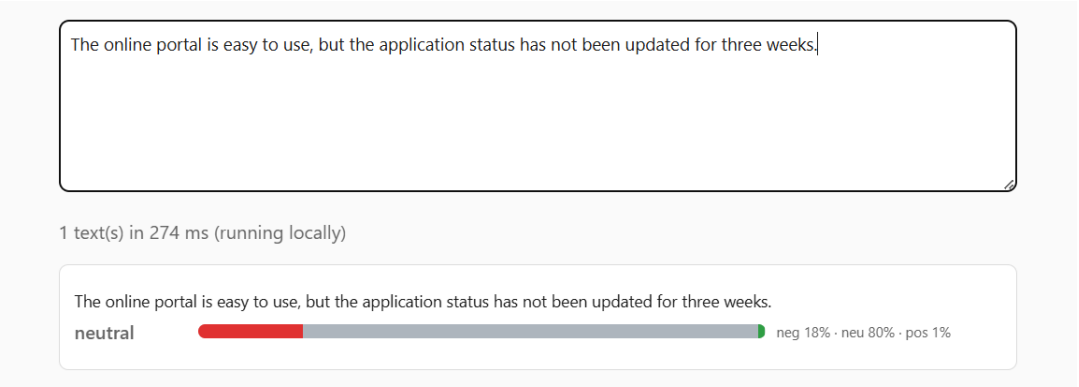


Figure 3.3: Sentiment-analysis result for English sample E3.

3.3 Marathi Examples

M1 input: मी उत्पन्नाच्या दाखल्यासाठी दोन महिन्यांपूर्वी अर्ज केला, पण अजूनही तो प्रलंबित आहे. हा विलंब मान्य नाही.

English meaning: I applied for an income certificate two months ago, but it is still pending. This delay is unacceptable. The model returned negative (96%), agreeing with the intended meaning.

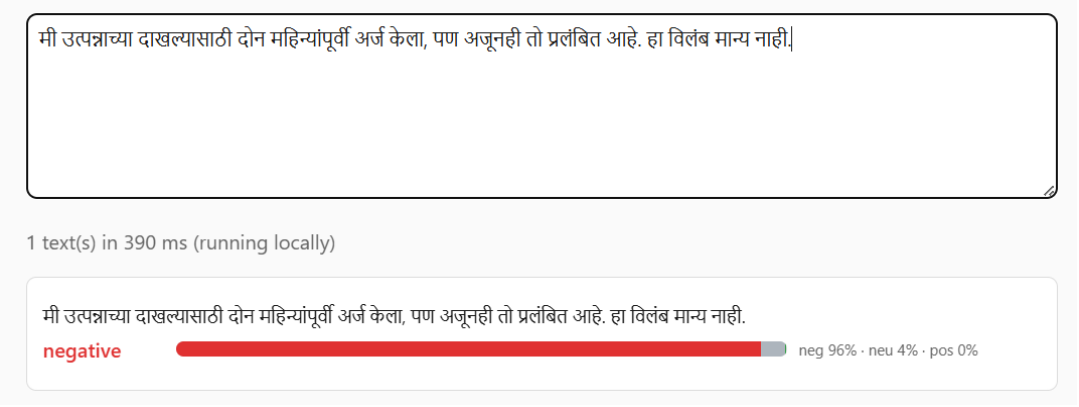


Figure 3.4: Sentiment-analysis result for Marathi sample M1.

M2 input: अधिकाऱ्यांनी माझ्या रेशन कार्डची समस्या लवकर सोडवली. चांगल्या सेवेबद्दल धन्यवाद.

English meaning: The officials resolved the issue with my ration card promptly. Thank you for the good service. The model returned positive (100%).

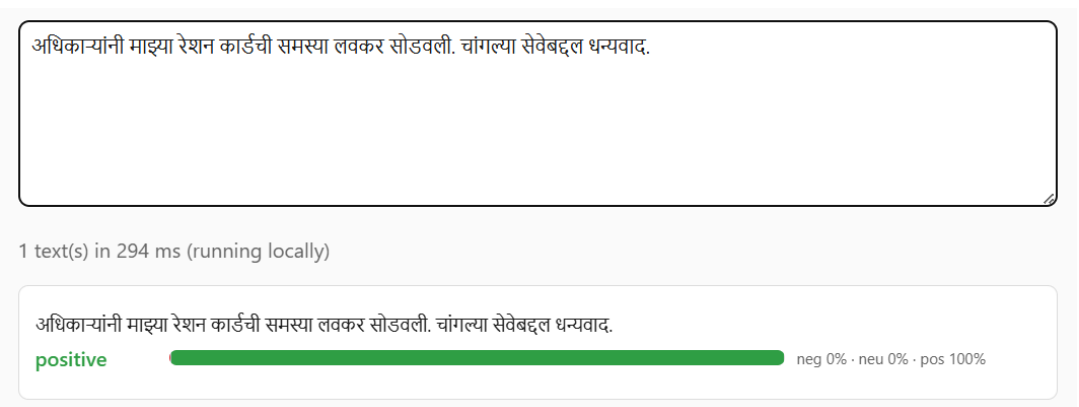


Figure 3.5: Sentiment-analysis result for Marathi sample M2.

M3 input: ऑनलाइन पोर्टल वापरण्यास सोपे आहे, पण तीन आठवड्यांपासून अर्जाची स्थिती अपडेट झालेली नाही.

English meaning: The online portal is easy to use, but the application status has not been updated for three weeks. The model returned negative (70%), emphasizing the unresolved status over the positive observation about usability.



Figure 3.6: Sentiment-analysis result for Marathi sample M3.

3.4 Machine Translation Results

Google Translate was tested with the three Marathi samples.

Table 3.2: Observed Marathi-to-English translations

ID	Observed translation summary	Assessment
M1	Income certificate application remains pending; delay is unacceptable.	Main complaint preserved.
M2	Ration card issue resolved promptly; thanks for good service.	Appreciation preserved.
M3	Portal is easy to use, but status has not been updated for three weeks.	Both clauses preserved.

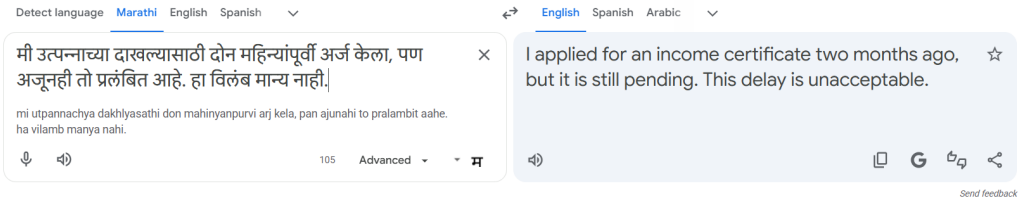


Figure 3.7: Google Translate output for Marathi sample M1.

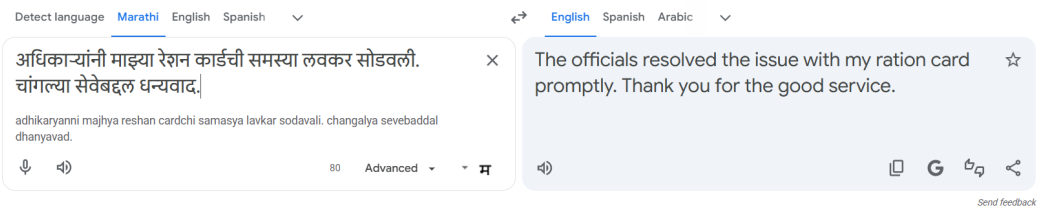


Figure 3.8: Google Translate output for Marathi sample M2.

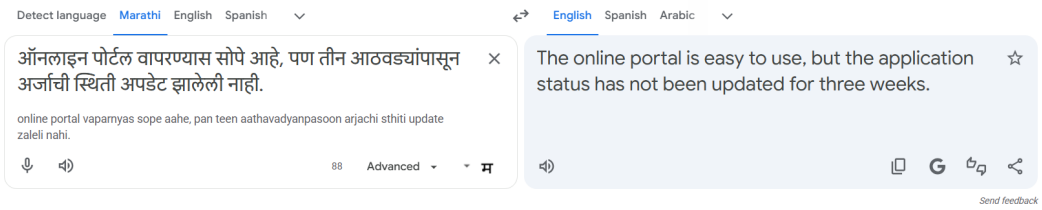


Figure 3.9: Google Translate output for Marathi sample M3.

Additional English-to-Marathi input: “The service is good, but the officer has not resolved my complaint yet.”

Observed output: सेवा चांगली आहे, परंतु अधिकाऱ्याने माझी तक्रार अजूनही सोडवलेली नाही.

The translation preserves the positive service description and the unresolved complaint.

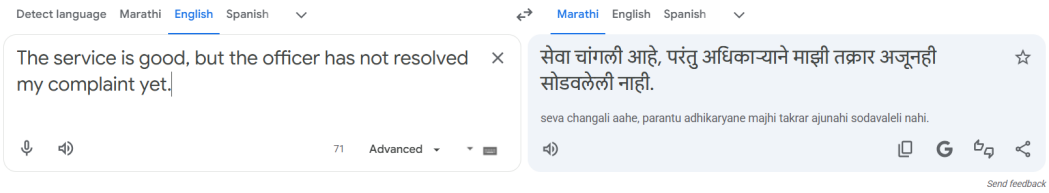


Figure 3.10: Google Translate output for the additional English-to-Marathi test.

4. Part II: Comparative and Error Analysis

4.1 Comparison of Tools

The sentiment model classifies sentiment, while Google Translate converts text between languages. They are complementary tools rather than direct substitutes.

Table 4.1: Comparison of observed tool behaviour

Criterion	Sentiment model	Google Translate
Main task	Assigns positive, negative, or neutral label.	Translates between languages.
English tests	Clear positive and negative examples matched intended meaning; E3 was neutral.	Not used as a sentiment classifier.
Marathi tests	M1 negative, M2 positive, M3 negative.	Main meaning preserved in the three tested examples.
Mixed feedback	One dominant label can hide contrasting opinions.	Tested translations retained both clauses.
Limitation	Does not explicitly represent each aspect's sentiment.	Quality assessed manually on a small sample only.

4.2 Error Matrix

Table 4.2: Error analysis matrix

Category	Evidence	Interpretation
Mixed sentiment	E3 was neutral; M3 was negative.	A single label does not explicitly capture both positive and negative aspects. This alone does not prove either label is incorrect.
Context representation	M3 received a 70% negative score despite praising portal usability.	The final label emphasizes the unresolved application status.
Translation fidelity	Main meaning was preserved in the three tested Marathi translations.	No major meaning loss was apparent in these examples; larger testing is needed.
Regional coverage	Only three Marathi sentiment samples were tested.	Dialects, informal spellings, and code-switching remain untested.
Cultural context	Samples concerned certificates and ration cards.	Sarcasm and region-specific cultural expressions were not directly tested.

4.3 Error Types and Limitations

Grammatical errors: No obvious grammatical error was noticed in the supplied translations, but this does not establish general grammatical accuracy.

Semantic errors: The tested translations retained the main meaning. More complex complaints may contain ambiguous references, idioms, or administrative terminology.

Context loss: A single sentiment label can hide differences between satisfaction with a website and dissatisfaction with the service outcome.

Regional misunderstanding: No specific cultural misunderstanding was demonstrated by these samples. Local expressions, abbreviations, and informal language require further testing.

4.4 Experimental Limitations

The dataset contains only six fictional sentiment examples. Displayed scores were not independently calibrated. The two tools perform different tasks, so a common accuracy score would be inappropriate. CPGRAMS's internal NLP system was not tested, and no statistical significance analysis was performed.

5. Part III: Adapting Feedback Analysis for Marathi

5.1 Why Marathi Adaptation Is Needed

A grievance-analysis system that supports Marathi can make feedback more accessible to citizens who prefer their regional language. Adaptation requires more than translating English training examples: the system must account for Marathi grammar, Devanagari text, dialects, administrative vocabulary, and mixed-language sentences.

5.2 Required Linguistic Resources

Marathi corpora: Use appropriately licensed Marathi text relevant to public services, applications, complaints, delays, and suggestions. Data should represent different regions and writing styles. Personal information must be removed or protected.

Sentiment-labelled data: Annotate examples as positive, negative, neutral, or mixed. Additional complaint categories may include delayed applications, ration card issues, certificate problems, portal difficulties, staff behaviour, and successful resolution. Multiple annotators should label a subset and resolve disagreements using documented guidelines.

Tokenization and morphology: Marathi text processing should handle Devanagari characters, punctuation, word boundaries, inflection, and spelling variation. Morphological analysis can help relate different forms of a word.

Part-of-speech tagging and named entity recognition: These tools can identify grammatical roles and entities such as locations, departments, and service names. Personal identifiers should be handled securely.

Marathi-capable language models: A multilingual model can be fine-tuned on domain-specific examples. Its performance should be compared with a baseline and evaluated across language styles before deployment.

5.3 Sociolinguistic Challenges

Code-switching: Citizens may mix Marathi and English, for example, माझा □□□□□□□□□□□□ अजून □□□□□□□□ आहे. The system should recognize English administrative words within Marathi syntax.

Dialect variation: Vocabulary and expressions differ across regions. Training only on formal Marathi may reduce performance on informal feedback. Regional examples should be included without treating one dialect as universal.

Script variation: Marathi is commonly written in Devanagari, but some users write it in Latin transliteration, such as `Majha arj ajun pending aahe.` Common transliterations and spelling variations should be handled carefully.

Administrative vocabulary: A domain glossary should cover certificate applications, ration cards, departments, status updates, and common equivalents in English and Marathi.

Sarcasm and indirect complaints: Literal positive words may express dissatisfaction in context. Carefully annotated examples are needed to test whether the system understands the intended meaning.

5.4 Proposed Adaptation Workflow

1. Collect authorized or appropriately licensed Marathi grievance text.

2. Remove or protect personal and sensitive information.
3. Normalize text while preserving meaning and language variation.
4. Annotate sentiment, complaint category, and relevant entities.
5. Split data into training, validation, and test sets to prevent leakage.
6. Train a baseline and fine-tune a suitable Marathi-capable model.
7. Evaluate formal Marathi, informal Marathi, code-switched text, and regional variations separately.
8. Review errors with Marathi speakers and domain experts.
9. Use human review for uncertain or high-impact cases.

5.5 Evaluation Metrics

Recommended metrics include precision, recall, macro-F1, and a confusion matrix for sentiment classification. Translation should be assessed by human reviewers for meaning preservation, terminology, and readability. Results should be reported separately for English, formal Marathi, informal Marathi, and code-switched text.

5.6 Recommended Improvements

1. Use aspect-based sentiment analysis to distinguish opinions about the portal, staff, and service outcome.
2. Allow mixed sentiment instead of forcing every sentence into one class.
3. Train and evaluate on a larger Marathi grievance dataset.
4. Include transliteration, code-switching, and regional vocabulary.
5. Provide human review for uncertain or high-impact complaints.
6. Monitor errors after deployment and update the model using governed data.

6. Conclusion

This assignment studied E-Governance Feedback Analysis using CPGRAMS as the real-world context. A sentiment-analysis demo and Google Translate were tested with fictional English and Marathi feedback. The model identified clear positive and negative examples but assigned a single dominant label to mixed feedback. Google Translate preserved the main meaning in the small set of tested examples.

The experiment did not directly evaluate CPGRAMS's internal NLP system and is too small to establish broad model accuracy. Adapting grievance analysis for Marathi requires domain-specific linguistic resources, coverage of regional and code-switched language, privacy safeguards, systematic evaluation, and human review. Aspect-based and mixed-sentiment analysis may better preserve the different concerns expressed by citizens.

Bibliography

[1] Government of India, *Centralised Public Grievance Redress and Monitoring System (CPGRAMS)*. Available: https://pgportal.gov.in/ (accessed October 2026).

[2] Government of Maharashtra, *Aaple Sarkar Grievance Redressal Portal*. Available: https://grievances.maharashtra.gov.in/ (accessed October 2026).

[3] Horizon-Labs, *Multilingual Sentiment Analysis Model Demonstration*. Available: https://huggingface.co/Horizon-Labs/multilingual-sentiment-small (accessed October 2026).

[4] Google, *Google Translate*. Available: https://translate.google.com/ (accessed October 2026).

[5] D. Jurafsky and J. H. Martin, *Speech and Language Processing*, online draft. Available: https://web.stanford.edu/~jurafsky/slp3/.